

Möbius inversion at the objective level

This is the last time I will be presenting at the SUSTech Graduate Topology Seminar during my master's studies. This note is a tribute to all the teachers and friends who have supported me along the way — we shared a wonderful time together, and I am grateful to you for witnessing my growth.

The convolution of the zeta function and the Möbius function returns to the origin, yet our lives move ever forward ☺

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This note is mainly written on a series of papers. For detailed research on decomposition spaces (used to construct incidence coalgebra in this note) and its application in combinatorics, one can read [2],[3],[4]; For the theory of Möbius inversion, one can refer to [5], and the most key point is summarised in the first section of this note; For the detailed version of cardinality (a tool transposes categorified theory to numbers), I recommend [1]; Besides, I also recommend a further reading about negative sets, which is a generalization of negative numbers, in [6].

1 Classical Möbius inversion

Construction 1.1. (Convolution algebra) Let (C, Δ, ε) be a coalgebra and (A, m, u) be an algebra. Denote

$$\Delta(x) = \sum_{(x)} x_{(1)} \otimes x_{(2)}.$$

Define the convolution product on $\text{Lin}(C, A)$ by

$$(\alpha * \beta)(x) = \sum_{(x)} m(\alpha(x_{(1)}), \beta(x_{(2)})).$$

Then $(\text{Lin}(C, A), *, u\varepsilon)$ forms an algebra, called the *convolution algebra*. □

Classical generalized Möbius inversion studies which elements of $\text{Lin}(C, A)$ are invertible for convolution.

Example 1.2. $(\mathbb{C}[\mathbb{N}^\times], \Delta, \varepsilon)$ is a coalgebra, where

$$\Delta(n) := \sum_{d|n} d \otimes \frac{n}{d}, \quad \varepsilon(n) = \begin{cases} 1, & n = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Here $\mathbb{C}$ is an algebra, and

$$\text{Lin}(\mathbb{C}[\mathbb{N}^\times], \mathbb{C}) = \{f : \mathbb{N}^\times \rightarrow \mathbb{C}\}.$$

A linear map is determined on the basis. In the following we also represent $\zeta, \mu \in \text{Lin}(\mathbb{C}[\mathbb{N}^\times], \mathbb{C})$ by maps $\mathbb{N}^\times \rightarrow \mathbb{C}$.

Consider

$$\zeta : \mathbb{N}^\times \rightarrow \mathbb{C}, \quad n \mapsto 1,$$

and

$$\mu : \mathbb{N}^\times \rightarrow \mathbb{C}, \quad n \mapsto \begin{cases} 0, & \text{if } n \text{ contains a square factor,} \\ (-1)^r, & \text{if } n \text{ is the product of } r \text{ distinct primes.} \end{cases}$$

Then $\zeta, \mu \in \text{Lin}(\mathbb{C}[\mathbb{N}^\times], \mathbb{C})$ and

$$\zeta * \mu = \mu * \zeta = \varepsilon.$$

□

Theorem 1.3. If $\phi \in \text{Lin}(C, A)$ sends all group-like elements to 1, then ϕ is convolution invertible. Its inverse ψ is given recursively by

$$\psi = e - \psi * (\phi - e), \quad e := u\varepsilon.$$

Proof sketch. Let

$$\psi_0 = e, \quad \psi_{n+1} := \psi_n * (\phi - e).$$

Define

$$\psi_{\text{even}} := \sum_{n \text{ even}} \psi_n, \quad \psi_{\text{odd}} := \sum_{n \text{ odd}} \psi_n,$$

and then set

$$\psi := \psi_{\text{even}} - \psi_{\text{odd}}.$$

□

We now formulate the above theorem at the objective level, i.e. categorify the classical generalized Möbius inversion theorem.

2 Categorified analogue of the coalgebra (C, Δ, ε)

Coalgebra is a vector space with compatible comultiplication vector space + comultiplication.

Notation. E, B are groupoids. X is a simplicial space, i.e., a functor $\Delta^{op} \rightarrow \mathcal{S}$, where $\mathcal{S}$ is the ∞ -category of spaces. Write $\ulcorner b \urcorner : 1 \rightarrow B$ for the map picking out b . For $A \in \text{Grpd}$, define

$$A^{\ulcorner b \urcorner} := \left[A \rightarrow 1 \xrightarrow{\ulcorner b \urcorner} B \right] \in \text{Grpd}_{/B}.$$

Homotopy fiber. Given $f : E \rightarrow B$ and $b \in B$, define the homotopy fiber E_b by the homotopy pullback square

$$\begin{array}{ccc} E_b & \longrightarrow & E \\ \downarrow & \lrcorner & \downarrow f \\ 1 & \xrightarrow{\lceil b \rceil} & B. \end{array}$$

Fact 2.1. For any $f : E \rightarrow B$ in $\mathbf{Grpd}_B$, $f \simeq \int^{b \in B} E_b \lceil b \rceil$. This suggests that $\mathbf{Grpd}_B$ behaves like a vector space:

- Basis: $\lceil b \rceil : 1 \rightarrow B \quad ([b] \in \pi_0 B)$
- Scalar multiplication: $A \lceil b \rceil$
- Addition: homotopy sum

□

Warning.

- 1) We have only defined “scalar multiplication” for $\lceil b \rceil : 1 \rightarrow B$, not arbitrary $E \rightarrow B$ in $\mathbf{Grpd}_B$.
- 2) There is no negative sign here; for example $-\lceil b \rceil$ is meaningless.

□

Anyway, we obtain a categorified analogue of a vector space. Next we define comultiplication and counit.

We want functors

$$\mathbf{Grpd}_B \rightarrow \mathbf{Grpd}_B \times \mathbf{Grpd}_B, \quad \mathbf{Grpd}_B \rightarrow \mathbf{Grpd}_1.$$

Step 1. Using

$$\mathbf{Grpd}_B \times \mathbf{Grpd}_B \cong \mathbf{Grpd}_{B \times B}, \quad \mathbf{Grpd}_1 \cong \mathbf{Grpd},$$

Step 2. A span

$$A \xleftarrow{f} C \xrightarrow{g} D$$

induces a functor

$$\mathbf{Grpd}_A \xrightarrow{f^*} \mathbf{Grpd}_C \xrightarrow{g_*} \mathbf{Grpd}_D.$$

$$\begin{array}{ccccccc} & & E' & \longrightarrow & E & & \\ & & \downarrow & & \downarrow & & \\ E \rightarrow A & \longrightarrow & C & \xrightarrow{f} & A & \longrightarrow & E' \rightarrow C \rightarrow D \end{array}$$

Remark 2.2. Since a classical coalgebra structure is bilinear, the functor $\mathbf{Grpd}_B \rightarrow \mathbf{Grpd}_{B \times B}$ should be represented by a span, which is the analogue of linear maps in category theory. □

Step 3. It is difficult to define a coalgebra structure on $\mathbf{Grpd}_B$ for arbitrary $B \in \mathbf{Grpd}$. Let $X : \Delta^{\text{op}} \rightarrow \mathcal{S}$ be a simplicial space.

By definition, X is an ∞ -groupoid. But in many cases we only use simplicial groupoids, viewing an ordinary groupoid as an ∞ -groupoid.

If $B = X_1$ for some simplicial space X , then we have a span

$$X_1 \xleftarrow{d_1} X_2 \xrightarrow{(d_2, d_0)} X_1 \times X_1 \rightsquigarrow \Delta : \mathrm{Grpd}_{/X_1} \rightarrow \mathrm{Grpd}_{/X_1 \times X_1}$$

and also a span

$$X_1 \xleftarrow{s_0} X_0 \rightarrow 1 \rightsquigarrow \varepsilon : \mathrm{Grpd}_{/X_1} \rightarrow \mathrm{Grpd}_{/1}$$

Step 4. We require Δ and ε to be coassociative and counital. These properties hold when X is a *decomposition space*.

The coalgebra $(\mathrm{Grpd}_{/X_1}, \Delta, \varepsilon)$ is called the *incidence coalgebra* of the decomposition space X .

Remark 2.3. (Coassociativity) We want the following square of slice groupoids to commute:

$$\begin{array}{ccc} \mathrm{Grpd}_{/X_1} & \xrightarrow{\Delta} & \mathrm{Grpd}_{/X_1 \times X_1} \\ \Delta \downarrow & & \downarrow \Delta \otimes \mathrm{id} \\ \mathrm{Grpd}_{/X_1 \times X_1} & \xrightarrow{\mathrm{id} \otimes \Delta} & \mathrm{Grpd}_{/X_1 \times X_1 \times X_1} \end{array}$$

At the level of spans, one is tempted to look at

$$\begin{array}{ccccc} X_1 & \xleftarrow{d_1} & X_2 & \xrightarrow{(d_2, d_0)} & X_1 \times X_1 \\ d_1 \uparrow & & & & \uparrow d_1 \times \mathrm{id} \\ X_2 & & & & X_2 \times X_1 \\ (d_2, d_0) \downarrow & & & & \downarrow (d_2, d_0) \times \mathrm{id} \\ X_1 \times X_1 & \xleftarrow{\mathrm{id} \times d_1} & X_1 \times X_2 & \xrightarrow{\mathrm{id} \times (d_2, d_0)} & X_1 \times X_1 \times X_1 \end{array}$$

However, in this diagram, the notion of commutativity is vacuous. □

Fact 2.4. For any homotopy pullback square

$$\begin{array}{ccc} H & \xrightarrow{q} & E \\ p \downarrow & \lrcorner & \downarrow g \\ G & \xrightarrow{f} & B, \end{array}$$

the functors

$$q!p^*, \quad g^*f! : \mathrm{Grpd}_{/G} \rightarrow \mathrm{Grpd}_{/E}$$

are naturally homotopy equivalent. □

Instead, decompose the problem into four smaller squares:

$$\begin{array}{ccccc} X_1 & \xleftarrow{d_1} & X_2 & \xrightarrow{(d_2, d_0)} & X_1 \times X_1 \\ d_1 \uparrow & (1) & d_1 \uparrow & (2) & \uparrow d_1 \times \mathrm{id} \\ X_2 & \xleftarrow{d_2} & X_3 & \xrightarrow{(d_3, d_0 d_0)} & X_2 \times X_1 \\ (d_2, d_0) \downarrow & (3) & \downarrow (d_2 d_2, d_0) & (4) & \downarrow (d_2, d_0) \times \mathrm{id} \\ X_1 \times X_1 & \xleftarrow{\mathrm{id} \times d_1} & X_1 \times X_2 & \xrightarrow{\mathrm{id} \times (d_2, d_0)} & X_1 \times X_1 \times X_1 \end{array}$$

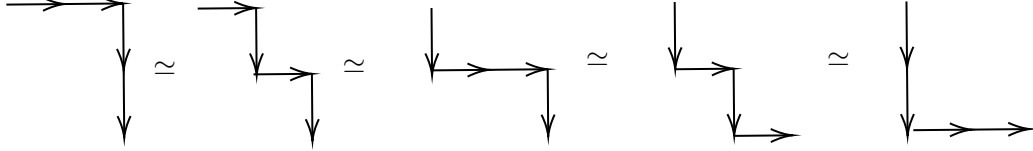
Squares (1) and (4) commute because simplicial object satisfying $d_i d_j = d_{j-1} d_i$ ($i < j$).

For example, square (4) commutes meaning that $(d_2 d_3 x, d_0 d_3 x, d_0 d_0 x) = (d_2 d_2 x, d_2 d_0 x, d_0 d_0 x)$ which is equivalent to $d_2 d_3 = d_2 d_2$ and $d_0 d_3 = d_2 d_0$.

Assume now that squares (2) and (3) are homotopy pullbacks. Then consider the morphism induced between slice groupoids.

$$\begin{array}{ccccc}
 \mathrm{Grpd}_{/X_1} & \xrightarrow{d_1^*} & \mathrm{Grpd}_{/X_2} & \xrightarrow{(d_2, d_0)!} & \mathrm{Grpd}_{/X_1 \times X_1} \\
 \downarrow & & \downarrow & & \downarrow \\
 \mathrm{Grpd}_{/X_2} & \xrightarrow{(1)'} & \mathrm{Grpd}_{/X_3} & \xrightarrow{(2)'} & \mathrm{Grpd}_{/X_2 \times X_1} \\
 \downarrow & & \downarrow & & \downarrow \\
 \mathrm{Grpd}_{/X_1 \times X_1} & \xrightarrow{(3)'} & \mathrm{Grpd}_{/X_1 \times X_2} & \xrightarrow{(4)'} & \mathrm{Grpd}_{/X_1 \times X_1 \times X_1}
 \end{array}$$

We have equivalences of paths in above diagram:



The first and fourth equivalence of paths is because (2)' and (3)' are pullbacks, respectively. The second and third equivalence of paths is because (1)' and (4)' commute, respectively.

This implies the original coassociativity square commutes.

So the axioms of a decomposition space should imply that the (2) and (3) are pullbacks.

□

3 Categorical analogues of (A, m, u) and $\mathrm{Lin}(C, A)$

Categorified algebra.

$$(\mathrm{Grpd}, \times, 1)$$

is an algebra, where $X \times Y$ is the Cartesian product of groupoids.

Categorified convolution algebra. Let $\mathbf{LIN}$ be the category of slice groupoids whose morphisms are induced by spans.

For $F, G \in \mathbf{LIN}(\mathrm{Grpd}_{/X_1}, \mathrm{Grpd})$, suppose F is induced by the span $X_1 \leftarrow M \rightarrow 1$ and G is induced by $X_1 \leftarrow N \rightarrow 1$. Define their convolution by the span

$$X_1 \leftarrow M * N \rightarrow 1,$$

which is defined as the composition of spans:

$$\begin{array}{ccccc}
 X_1 & & & & \\
 \uparrow d_1 & \swarrow & & & \\
 X_2 & \xleftarrow{\quad} & M * N & & \\
 \downarrow (d_2, d_0) & & \downarrow & \searrow & \\
 X_1 \times X_1 & \xleftarrow{\quad} & M \times N & \longrightarrow & 1
 \end{array}$$

The neutral functor for convolution is

$$\varepsilon : \text{Grpd}_{/X_1} \rightarrow \text{Grpd}.$$

Fact 3.1. $\text{Grpd}_{/B} \simeq \text{Grpd}^B$ □

Hence

$$\mathbf{LIN}(\text{Grpd}_{/X_1}, \text{Grpd}) \simeq \text{Grpd}^{X_1}.$$

4 Cardinality

Taking cardinality turns the objective-level theory into the classical one.

Definition 4.1. If X is a finite groupoid, define the cardinality of X be $|X| := \sum_{[x] \in \pi_0 X} \frac{1}{|\text{Aut}(x)|} \in \mathbb{Q}$.

Property 4.2. Let A, B, G be locally finite groupoids and let

$$A \leftarrow G \rightarrow B$$

be a finite span. Then the induced functor $\text{Grpd}_{/A} \rightarrow \text{Grpd}_{/B}$ restricts to $\text{grp}_{/A} \rightarrow \text{grp}_{/B}$.

Let $\mathbf{lin}$ be the category of slice categories $\text{grp}_{/A}$, with A locally finite. There is a functor

$$\| \cdot \| : \mathbf{lin} \rightarrow \mathbf{Vect}$$

$$\begin{array}{ccc} \text{grp}_{/A} & \xrightarrow{\quad} & \mathbb{Q}^{\pi_0 A} \\ \downarrow & & \downarrow \\ \text{grp}_{/B} & \xrightarrow{\quad} & \mathbb{Q}^{\pi_0 B} \end{array} \quad \begin{array}{c} \delta_a \\ \downarrow \\ \sum_{[b] \in \pi_0 B} |B[b]| |G_{a,b}| \delta_b \end{array}$$

Here $\mathbb{Q}^{\pi_0 A} := \mathbb{Q}[\pi_0 A]$ is the vector space with basis $\{\delta_a\}_{[a] \in \pi_0 A}$. Dually there is a functor

$$\| \cdot \| : \text{grp}^B \rightarrow \text{grp}^A$$

$$\begin{array}{ccc} \text{grp}^B & \xrightarrow{\quad} & \mathbb{Q}^{\pi_0 B} \\ \downarrow & & \downarrow \\ \text{grp}^A & \xrightarrow{\quad} & \mathbb{Q}^{\pi_0 A} \end{array} \quad \begin{array}{c} \delta^b \\ \downarrow \\ \sum_{[a] \in \pi_0 A} |B[b]| |G_{a,b}| \delta^a \end{array}$$

Here $\mathbb{Q}^{\pi_0 B}$ is the function space, i.e. the profinite-dimensional vector space with pro-basis given by the characteristic functions δ^b .

Remark 4.3. One can think of

$$\mathbb{Q}^{\pi_0 B} = \varprojlim_{J \subset \pi_0 B, J \text{ finite}} \mathbb{Q}^J,$$

each $\mathbb{Q}^J$ finite dimensional. So a vector in $\mathbb{Q}^{\pi_0 B}$ can be represented by combination of probasis with infinite terms of nonzero coefficients. □

5 Möbius inversion at the objective level

We want to define $\zeta : \text{Grpd}_{/X_1} \rightarrow \text{Grpd}$. Suppose ζ is represented by a span $X_1 \xleftarrow{f} G \rightarrow 1$. A finite span $A \leftarrow G \rightarrow B$ induces $\text{Grpd}_{/A} \rightarrow \text{Grpd}_{/B}$. When A, B, G are locally finite, this restricts to $\text{grpd}_{/A} \rightarrow \text{grpd}_{/B}$.

Take cardinality, we obtain

$$\mathbb{Q}_{\pi_0 A} \rightarrow \mathbb{Q}_{\pi_0 B}, \quad \delta_a \mapsto \sum_{[b] \in \pi_0 B} |B_{[b]}| |G_{a,b}| \delta_b,$$

where $B_{[b]}$ is the full subgroupoid of B on objects isomorphic to b , and $G_{a,b}$ is the fiber of $G \rightarrow A \times B$ over (a, b) .

Classically, ζ is the constant function 1, we hope after taking cardinality, ζ becomes

$$\mathbb{Q}_{\pi_0 X_1} \rightarrow \mathbb{Q}, \quad \delta_x \mapsto 1.$$

Now let's find what ζ should be. Suppose ζ is given by $X_1 \xleftarrow{f} G \rightarrow 1$.

$$\begin{aligned} |\zeta| : \mathbb{Q}_{\pi_0 X_1} &\rightarrow \mathbb{Q}, \\ \delta_x &\mapsto \sum_{[b] \in \pi_0 1} |1_{[b]}| |G_{x,b}| \delta_b = |G_{x,*}| \delta_*. \end{aligned}$$

Remark 5.1. $1_{[*]}$ is the full subcategory of 1 consisting of objects isomorphic to $*$. So $|1_{[*]}| = 1$ $\square$

Hence we need to construct G and f , such that

$$|G_{x,*}| = 1 \quad \text{for all } [x] \in \pi_0 X_1.$$

If you thought of $G \rightarrow X_1 \times 1$ as morphism between sets and $|G_{x,*}|$ be the cardinality of the preimage set of $(x, *)$, the cheapest map to realize this is $X_1 \xleftarrow{\text{id}} X_1 \rightarrow 1$

$G_{x,*}$ is the pullback

$$\begin{array}{ccc} G_{x,*} & \longrightarrow & X_1 \\ \downarrow & \lrcorner & \downarrow \\ 1 & \xrightarrow{(x,*)} & X_1 \times 1. \end{array}$$

For two objects $(*, y, \phi : x \rightarrow y)$, $(*, y', \phi' : x \rightarrow y') \in G_{x,*}$, there is a morphism $(\text{id}, \phi' \phi^{-1}) : (*, y, \phi) \rightarrow (*, y', \phi')$.

Since we have commutative diagram:

$$\begin{array}{ccc} (x, *) & \xrightarrow{\text{id}} & (x, *) \\ \phi \downarrow & & \downarrow \phi' \\ (y, *) & \xrightarrow{\phi' \phi^{-1}} & (y', *) \end{array}$$

$\pi_0 G_{x,*}$ is a singleton.

Let $(*, y, \phi) \xrightarrow{(\text{id}, g)} (*, y, \phi)$ be an automorphism, then the above commutative diagram forces $g = \text{id}_y$. Therefore $|\text{Aut}(*, y, \phi)| = 1$. Hence $|G_{x,*}| = \frac{1}{|\text{Aut}(*, y, \phi)|} = 1$.

So the objective-level zeta functor is represented by

$$X_1 \xleftarrow{\text{id}} X_1 \rightarrow 1.$$

Towards an objective-level inverse

Goal: Find convolution inverse of ζ .

Idea and problem: In the classical case, we construct a family of $\{\phi_n\}$ and let

$$\mu = \sum_{n \text{ even}} \phi_n - \sum_{n \text{ odd}} \phi_n.$$

So $\zeta * (\phi_{\text{even}} - \phi_{\text{odd}}) = \varepsilon$. Note that in our case, ϕ_{odd} lies in $\text{Grpd}^{/X_1}$, where the “ $-\phi_{\text{odd}}$ ” does not make sense. So the right formula suggests that the Möbius inversion principle should be

$$\zeta * \phi_{\text{even}} = \varepsilon + \zeta * \phi_{\text{odd}},$$

which is sign free.

In the classical case one constructs a family $\{\phi_n\}$ and lets

$$\mu = \sum_{n \text{ even}} \phi_n - \sum_{n \text{ odd}} \phi_n,$$

so that

$$\zeta * (\phi_{\text{even}} - \phi_{\text{odd}}) = \varepsilon.$$

At the objective level, however, a negative sign does not make sense inside Grpd^{X_1} . So the correct sign-free formulation is

$$\zeta * \Phi_{\text{even}} = \varepsilon + \zeta * \Phi_{\text{odd}}.$$

Definition of Φ_n . There is a canonical map $X_n \rightarrow X_1$ obtained by composing all arrows:

$$X_n \xrightarrow{d_{i_n}} X_{n-1} \xrightarrow{d_{i_{n-1}}} X_{n-2} \rightarrow \cdots \rightarrow X_2 \xrightarrow{d_{i_2}} X_1,$$

where each d_{i_j} is an inner face map (since we only consider composition of arrows), so $1 \leq i_j \leq j-1$.

Claim: $d_{i_2}d_{i_3} \cdots d_{i_n} = d_1d_1 \cdots d_1$.

Indeed, one repeatedly uses

$$d_{j-1}d_i = d_id_j$$

to move d_j leftward until its index drops to 1. This can always be achieved. There are $j-2$ maps preceding d_{i_j} . Since $1 \leq i_j \leq j-1$, the maximum value i_j can take is $j-1$. By commuting d_{i_j} to the left (moving it forward), one can ultimately shift it past $j-2$ maps to reduce the index by $j-2$; that is, $d_{j-1-(j-2)} = d_1$. For example,

$$X_5 \rightarrow X_1, \quad d_2d_1d_3d_4 = d_1d_2d_1d_3 = d_1d_1d_2d_1 = d_1d_1d_1d_1.$$

6 Completeness and nondegenerate simplices

Definition 6.1. A decomposition space X is *complete* if $s_0 : X_0 \rightarrow X_1$ is monomorphism, i.e., each homotopy fiber of s_0 is either empty or contractible. $\square$

Example 6.2. Let G be a nontrivial group, and let X be the simplicial groupoid with $X_n = (BG)^n$. Then $s_0 : 1 \rightarrow BG$ is not monomorphism, so X is not complete.

Indeed, there is only one obj in BG and the only fiber groupoid is

$$\begin{cases} \text{objects:} & (*, *, g \in G) \\ \text{morphisms:} & \text{Hom}((*, *, g_1), (*, *, g_2)) = \begin{cases} (\text{id}, \text{id}) & g_1 = g_2 \\ \emptyset & \text{o/w} \end{cases} \end{cases}$$

, which can be computed by

$$\begin{array}{ccc} \text{fiber} & \longrightarrow & 1 \\ \downarrow & \lrcorner & \downarrow s_0 \\ 1 & \xrightarrow{\lceil x \rceil} & BG \end{array}$$

The diagram $\begin{array}{ccc} * & \xrightarrow{\text{id}} & * \\ g_1 \downarrow & & \downarrow g_2 \\ * & \xrightarrow{\text{id}} & * \end{array}$ commutes iff $g_1 = g_2$. Note $\text{Hom}_1(*, *) = \{\text{id}\}$

This fiber groupoid is discrete but not connected, so it's not contractible.

So s_0 is not a monomorphism.

Another argument: if s_0 were mono, then s_0 would be fully faithful, so $s_0(1)$ would be the full subgroupoid of BG . That would force $\text{Hom}_1(*, *) \simeq \text{Hom}_{BG}(*, *)$, impossible when G is nontrivial. $\square$

Motivation for completeness

We want to define nondegenerate simplices for a simplicial groupoid X .

For simplicial sets, the set of nondegenerate simplices is the complement of the image of degeneracy maps. For simplicial groupoids, one would like the complement of the image *groupoid*. But in general that complement need not itself be a groupoid.

The “complement” of the image groupoids in general is not a groupoid, for e.g., let X be a category with one object and only one nontrivial morphism e with $e^2 = \text{id}$. Consider $1 \rightarrow X$. The complement is a 1-mor e , does not contain any objects. So the complement is not a groupoid.

So to make the complement of the image meaningful, we want the degeneracy map to be fully faithful, i.e. a monomorphism.

Fact 6.3. In a decomposition space, a simplex is nondegenerate if and only if all its principal edges are nondegenerate. For example, in a 2-simplex the principal edges are x_{01} and x_{12} , not x_{02} . In a 3-simplex the principal edges are x_{01}, x_{12}, x_{23} .

Hence it suffices to make nondegenerate 1-simplices make sense, i.e. it suffices to make the complement of $s_0 : X_0 \rightarrow X_1$ makes sense. This is why the condition of completeness is needed. $\square$

Constructing Φ_n

Classical situation

$$\Phi_0 = \varepsilon, \quad \Phi_1 = \zeta - \varepsilon, \quad \Phi_n = \Phi_{n-1} * (\zeta - \varepsilon),$$

hence, $\Phi_n = (\zeta - \varepsilon)^n = \Phi_1^n$.

Objective level

Define $\vec{X}_n \subset X_n$ to be the full subgroupoid of nondegenerate n -simplices. ε is induced by span Φ_0 is represented by $X_1 \xleftarrow{s_0} X_0 \rightarrow 1$, and ζ is induced by span $X_1 \xleftarrow{\text{id}} X_1 \rightarrow 1$. Hence it's reasonable to define $\Phi_1 = “\zeta - \varepsilon”$ induced by $X_1 \leftarrow \vec{X}_1 \rightarrow 1$. More generally, define Φ_n induced by span $X_1 \leftarrow \vec{X}_n \rightarrow 1$ where $\vec{X}_0 = X_0$ and $\vec{X}_n \rightarrow X_1$ is the restriction of the canonical morphism $X_n \rightarrow X_1$.

The next proposition illustrate our Φ_n have same behavior as classical case.

Property 6.4. $\Phi_n = (\Phi_1)^n$

Proof sketch. We prove by induction. Assume $\Phi_{n-1} = (\Phi_1)^{n-1}$. Then $(\Phi_1)^n = \Phi_{n-1} * \Phi_1$, which is given by

$$\begin{array}{ccc} & X_1 & \\ \uparrow & & \\ X_2 & \xleftarrow{\quad} & ? \\ \downarrow & \lrcorner & \downarrow \\ X_1 \times X_1 & \xleftarrow{\quad} & \vec{X}_{n-1} \times \vec{X}_1 \longrightarrow 1 \end{array}$$

Using [Lemma 3.5, [3]] one can show $? = X_{a\dots a} = \vec{X}_n$, which shows $(\Phi_1)^n = \Phi_n$.

7 Sign-free Möbius inversion for complete decomposition spaces

Lemma 7.1. The square

$$\begin{array}{ccc} \vec{X}_1 + \vec{X}_2 & \longrightarrow & X_2 \\ \downarrow & \lrcorner & \downarrow (d_2, d_0) \\ X_1 \times \vec{X}_1 & \longrightarrow & X_1 \times X_1 \end{array}$$

is a pullback.

Proof. (not strict but gives a feeling of how it works) An object in the pullback is a tuple (a, b, σ, φ) with

$$(a, b) \xrightarrow{\varphi} (d_2\sigma, d_0\sigma).$$

Case 1. If a is degenerate, then σ is determined by $b \in \vec{X}_1$ (up to equivalence)

$$\begin{array}{ccc} & \bullet & \\ \text{id} \simeq d_2\sigma \nearrow & & \searrow d_0\sigma \simeq b \\ \bullet & \xrightarrow{d_1\sigma} & \bullet \end{array}$$

So (a, b, σ, ϕ) can be encoded in $b \in \vec{X}_1$ (up to equivalence)

Case 2. If a is nondegenerate, then (a, b, σ, ϕ) is encoded by a nondegenerate 2-simplex $\sigma \in \vec{X}_2$, whose principal edges are $d_2\sigma \simeq a$, $d_0\sigma \simeq b$, both are nondegenerate. For a complete decomposition space, nondegeneracy of the principal edges implies nondegeneracy of the whole simplex, indeed $\sigma \in \vec{X}_2$.

Therefore,

$$\begin{array}{ccc} d_1\sigma & & X_1 \\ \uparrow & & \uparrow \\ \sigma & & X_2 \\ \downarrow & & \downarrow \\ (d_2\sigma, d_0\sigma) & & X_1 \times X_1 \xleftarrow{\quad} \vec{X}_1 + \vec{X}_1 \xrightarrow{\quad} 1 \\ & \searrow \phi \simeq & \\ & & (a, b) \xleftarrow{\quad} (a, b) \end{array}$$

Case I: If a is degenerate, (a, b, σ, φ) encoded by $b \in \vec{X}_1$, and in this case $d_1\sigma \simeq b$. $\left(\begin{array}{c} \vec{X}_1 \rightarrow X_1 \\ b \mapsto b \end{array} \right)$

Case II: If a is nondegenerate, (a, b, σ, φ) encoded by $\sigma \in \vec{X}_2$, and in this case $d_1\sigma \simeq b \circ a$. $\left(\begin{array}{c} \vec{X}_2 \rightarrow X_1 \\ \sigma \mapsto d_1\sigma \end{array} \right)$

Hence, $\zeta * \Phi_1 \simeq \Phi_1 + \Phi_2$. Higher-dimensional analogues gives

$$\zeta * \Phi_n \simeq \Phi_n + \Phi_{n+1}.$$

□

Remark 7.2. For a rigorous version, there is a proof using “word notation”. See **Section 2.8**, **Lemma 2.12**, **Proposition 2.13**, **Proposition 3.7** in [3].

Theorem. For a complete decomposition space,

$$\zeta * \Phi_{\text{even}} = \varepsilon + \zeta * \Phi_{\text{odd}}, \quad \Phi_{\text{even}} * \zeta = \varepsilon + \Phi_{\text{odd}} * \zeta,$$

where

$$\Phi_{\text{even}} := \sum_{k \geq 0} \Phi_{2k}, \quad \Phi_{\text{odd}} := \sum_{k \geq 0} \Phi_{2k+1}.$$

Consequently,

$$\begin{aligned} \zeta * \Phi_{\text{even}} &= \zeta * \sum_{k \geq 0} \Phi_{2k} = \sum_{k \geq 0} \zeta * \Phi_{2k} = \sum_{k \geq 0} (\Phi_{2k} + \Phi_{2k+1}) = \sum_{k \geq 0} \Phi_k, \\ \zeta * \Phi_{\text{odd}} + \varepsilon &= \zeta * \left(\sum_{k \geq 0} \Phi_{2k+1} \right) + \Phi_0 = \sum_{k \geq 0} (\Phi_{2k+1} + \Phi_{2k+2}) + \Phi_0 = \sum_{k \geq 0} \Phi_k. \end{aligned}$$

Hence

$$\zeta * \Phi_{\text{even}} = \zeta * \Phi_{\text{odd}} + \varepsilon.$$

Similarly,

$$\Phi_{\text{even}} * \zeta = \Phi_{\text{odd}} * \zeta + \varepsilon.$$

Theorem 7.3. If X is a Möbius decomposition space, then

$$|\zeta| * |\mu| = |\varepsilon| = |\mu| * |\zeta|,$$

where

$$|\mu| := |\Phi_{\text{even}}| - |\Phi_{\text{odd}}|.$$

8 Cancellation example: $\mathbf{N} := N((\mathbb{N}, +))$

$(\mathbb{N}, +)$ is a monoid and $B(\mathbb{N}, +)$ is the category with one object with hom set be $(\mathbb{N}, +)$.

Let $\mathbf{N}$ be the nerve of the monoid $(\mathbb{N}, +)$.

The objective-level Φ_r . Φ_r is represented by the span

$$\mathbf{N}_1 \leftarrow \vec{\mathbf{N}}_r \rightarrow 1.$$

Here

$$\mathbf{N}_r = N(B\mathbb{N}) = \{(n_1, \dots, n_r) \mid n_i \in \mathbb{N}\},$$

and the nondegenerate part is

$$\vec{\mathbf{N}}_r = \{(n_1, \dots, n_r) \mid n_i \in \mathbb{N} \setminus \{0\}\} = (\mathbb{N} \setminus \{0\})^r.$$

The fiber of

$$\vec{\mathbf{N}}_r \rightarrow \mathbf{N}_1 = \mathbb{N}, \quad (n_1, \dots, n_r) \mapsto n_1 + \dots + n_r$$

over $n \in \mathbb{N}$ is

$$(\vec{\mathbf{N}}_r)_n = \{(n_1, \dots, n_r) \in (\mathbb{N} \setminus \{0\})^r \mid n_1 + \dots + n_r = n\}.$$

Since $\mathbb{N}$ is a set, these fibers are just sets, namely the set of r -partitions of n :

$$(\vec{\mathbf{N}}_r)_n = \{n \twoheadrightarrow r\}.$$

Therefore

$$\Phi_r(\ulcorner n \urcorner) = \{n \twoheadrightarrow r\}.$$

Hence

$$\Phi_{\text{even}}(n) = \{n \twoheadrightarrow r \mid r \text{ even}\}, \quad \Phi_{\text{odd}}(n) = \{n \twoheadrightarrow r \mid r \text{ odd}\}.$$

With an abusive sign notation one would like to write

$$\mu(n) = \sum_{r \geq 0} (-1)^r \{n \twoheadrightarrow r\},$$

but this only makes sense after taking cardinality.

Remark 8.1. Note that here we use “negative sets”, which does not make sense.

Cardinality computation.

$$|\{n \twoheadrightarrow r\}| = \binom{n-1}{r-1}.$$

Remark 8.2. Since each cell must contain at least one element, the problem reduces to choosing $r-1$ positions for the bars from the $n-1$ available gaps between elements.

Thus

$$\mu(n) = \sum_{r \geq 0} (-1)^r \binom{n-1}{r-1}.$$

Writing

$$m := n-1, \quad s := r-1,$$

this becomes

$$\mu(n) = \sum_{s \geq -1} (-1)^{s+1} \binom{m}{s}.$$

For $n=1$ one gets

$$\mu(1) = (-1)^0 \binom{0}{-1} + (-1)^1 \binom{0}{0} = -1,$$

$$\text{For } n=0 \text{ one gets } \mu(0) = \underbrace{(-1)^0 \binom{-1}{-1}}_1 + \underbrace{\binom{-1}{0} + \binom{-1}{1} + \dots}_0 = 1$$

For $n > 1$,

$$\mu(n) = \sum_{s=0}^m (-1)^{s+1} \binom{m}{s} = -(x-1)^m|_{x=1} = 0.$$

Hence

$$\mu(n) = \begin{cases} 1, & n = 0, \\ -1, & n = 1, \\ 0, & \text{otherwise.} \end{cases}$$

So, in $\mathbb{Q}^{\pi_0 \mathbb{N}} \cong \mathbb{Q}^{\mathbb{N}}$,

$$\mu = \sum \mu(n) \delta^n = \delta^0 - \delta^1.$$

Another way to see this

Under cardinality,

$$\text{grpd}^{\mathbb{N}} \rightsquigarrow \mathbb{Q}^{\pi_0 \mathbb{N}_1} = \mathbb{Q}^{\mathbb{N}},$$

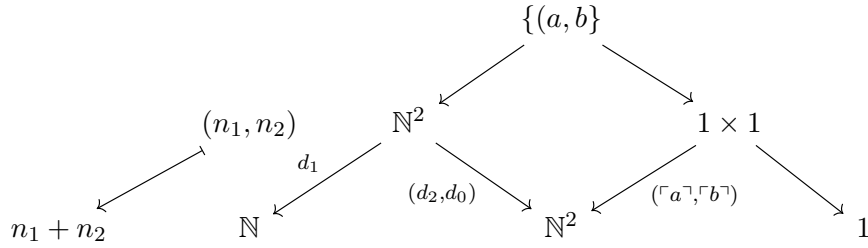
The incidence algebra

$$(\text{grpd}^{\mathbb{N}}, *, \varepsilon)$$

has basis $\{h^n\}_{n \in \mathbb{N}}$ with convolution $h^a * h^b = h^{a+b}$.

grpd^S is the full subcategory of $\text{Grpd}^S \simeq \mathbf{LIN}(\text{Grpd}_S, \text{Grpd})$. So an object in grpd^S is a linear functor $\text{Grpd}_S \rightarrow \text{Grpd}$. Since functor is linear, it's induced by span $S \leftarrow G \rightarrow 1$.

The basis $h^s \in \text{grpd}^S$ is induced by $S \xleftarrow{\lceil s \rceil} 1 \rightarrow 1$.



$h^a * h^b$ is given by $\mathbb{N} \leftarrow \{(a, b)\} \rightarrow 1$, $a + b \leftarrow (a, b)$, which is $\mathbb{N} \xleftarrow{\lceil a+b \rceil} 1 \rightarrow 1$, i.e., h^{a+b}

Therefore, after taking cardinality, the incidence algebra is $I^{\mathbb{N}}$.

$$I^{\mathbb{N}} \cong \mathbb{Q}[[z]], \quad |h^n| \longleftrightarrow z^n.$$

More explicitly,

$$f : \mathbb{N} \rightarrow \mathbb{Q} \quad \longleftrightarrow \quad \sum_{n \geq 0} f(n) z^n.$$

Here $f : \mathbb{N} \rightarrow \mathbb{Q}$ represents a vector in $I^{\mathbb{N}}$ whose coefficients of δ^n is $f(n)$.

The convolution law

$$\delta_a * \delta_b = \delta_{a+b}$$

corresponds to

$$z^a \cdot z^b = z^{a+b}.$$

In $\mathbb{Q}[[z]]$, since $f(n) \equiv 1$,

$$\zeta = \sum_{n \geq 0} z^n = \frac{1}{1-z},$$

so its inverse is

$$\mu = 1 - z.$$

This is exactly the function

$$0 \mapsto 1, \quad 1 \mapsto -1, \quad n \mapsto 0 \ (n \geq 2),$$

that is,

$$\mu = 1 \cdot \delta_0 + (-1) \cdot \delta_1 = \delta_0 - \delta_1 \text{ in } I^{\mathbb{N}}$$

9 Objective-level cancellation for $\mathbb{N}$

At the objective level, the vanishing for $n \geq 2$ comes from an actual cancellation of groupoids.

Bijection between odd and even decompositions. We denote an r -partition of n elements in $\Phi_r(n)$ as $(x_1, \dots, x_r)$ fullfilling $\sum_i x_i = n$.

For $n \geq 2$, there is a bijection

$$\Phi_{\text{odd}}(n) \longrightarrow \Phi_{\text{even}}(n)$$

by

$$(x_1, \dots, x_r) \mapsto \begin{cases} (1, x_1 - 1, x_2, \dots, x_r), & \text{if } x_1 > 1, \\ (x_2 + 1, x_3, \dots, x_r), & \text{if } x_1 = 1. \end{cases}$$

Remark 9.1. This works for $n \geq 2$, since $\Phi_{\text{even}}(1) = \emptyset$. □

Therefore

$$|\Phi_{\text{even}}(n)| = |\Phi_{\text{odd}}(n)|, \quad n \geq 2,$$

and so

$$\mu(n) = 0, \quad n \geq 2.$$

Checking the Möbius principle at the objective level for $\mathbb{N}$

We want to check

$$\zeta * \Phi_{\text{even}} = \zeta * \Phi_{\text{odd}} + \varepsilon.$$

The counit ε is represented by $\mathbf{N}_1 \xleftarrow{s_0} \mathbf{N}_0 \rightarrow 1$, $0 \leftarrow *$, since $\text{id}_* = 0 \in \mathbb{N}$ is the identity under composition in $B\mathbb{N}$.

This induces $\Phi_0 : \text{Grpd}_{/\mathbf{N}_1} \longrightarrow \text{Grpd}_{/\mathbf{N}_0} \longrightarrow \text{Grpd}_{/1} \simeq \text{Grpd}$

$$\begin{array}{ccccc} & ? & \longrightarrow & 1 & \\ & \downarrow & & \downarrow \lceil n \rceil & \\ 1 & \xrightarrow{\lceil n \rceil} & \mathbb{N} & \longmapsto & * \longrightarrow \mathbb{N} \longmapsto \begin{cases} \emptyset & \text{when } n \neq 0 \\ * & \text{when } n = 0 \end{cases} \end{array}$$

Hence

$$\varepsilon(\lceil n \rceil) = \begin{cases} \emptyset, & n \neq 0, \\ 1, & n = 0. \end{cases}$$

For $n \geq 2$, since $\Phi_{\text{even}}(n) \simeq \Phi_{\text{odd}}(n)$ and $\varepsilon(\lceil n \rceil) = \emptyset$, the identity

$$\zeta * \Phi_{\text{even}} = \zeta * \Phi_{\text{odd}} + \varepsilon$$

holds automatically.

So it remains to check the cases $n \leq 1$.

For $n \leq 1$ one has

$$\Phi_{\text{even}}(n) = \Phi_0(n), \quad \Phi_{\text{odd}}(n) = \Phi_1(n).$$

Therefore it suffices to show

$$\zeta * \Phi_0 = \zeta * \Phi_1 + \varepsilon.$$

Recall:

- Φ_r is given by $\mathbb{N} \leftarrow (\mathbb{N} \setminus \{0\})^r \rightarrow 1$, $x_1 + \cdots + x_r \leftarrow (x_1, \dots, x_r)$,
- Φ_0 is given by $\mathbb{N} \leftarrow * \rightarrow 1$, $0 \leftarrow *$,
- Φ_1 is given by $\mathbb{N} \leftarrow \mathbb{N} \setminus \{0\} \rightarrow 1$, $n \leftarrow n$,
- ζ is given by $\mathbb{N} \xleftarrow{\text{id}} \mathbb{N} \rightarrow 1$.

For $n \leq 1$, $\Phi_1(\ulcorner n \urcorner)$ coincides with $\delta^1(\ulcorner n \urcorner)$ of the element 1:

$\Phi_1 : \text{Grpd}_{/\mathbb{N}} \rightarrow \text{Grpd}$ is computed by

$$\begin{array}{c} \begin{array}{ccccc} & & ? & \longrightarrow & 1 \\ & & \downarrow & & \downarrow \ulcorner n \urcorner \\ 1 & \xrightarrow{\ulcorner n \urcorner} & \mathbb{N} & \longrightarrow & \mathbb{N} \setminus 0 & \longrightarrow & \mathbb{N} & \longrightarrow & \end{array} \\ \left\{ \begin{array}{ll} \emptyset & , n = 0 \\ \{n\} & , \text{otherwise} \end{array} \right. \end{array}$$

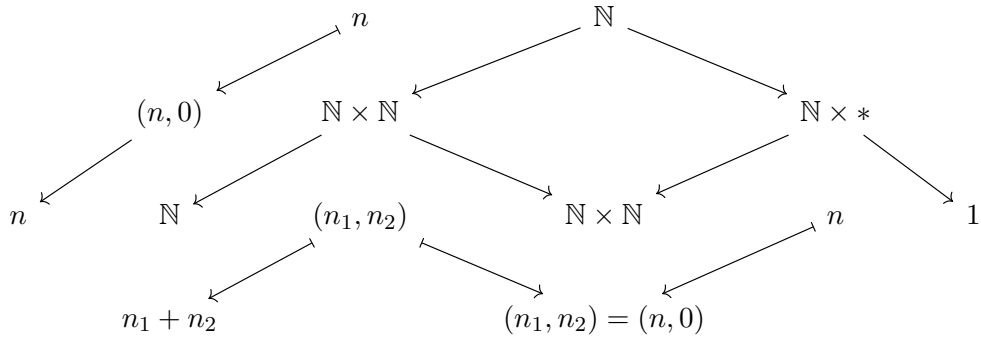
$\delta^1 : \text{Grpd}_{/\mathbb{N}} \rightarrow \text{Grpd}$ is computed by:

$$\begin{array}{c} \begin{array}{ccccc} & & ? & \longrightarrow & 1 \\ & & \downarrow & & \downarrow \ulcorner n \urcorner \\ 1 & \xrightarrow{\ulcorner n \urcorner} & \mathbb{N} & \longrightarrow & * & \longrightarrow & \mathbb{N} & \longrightarrow & \end{array} \\ \left\{ \begin{array}{ll} 1, & n = 1 \\ \emptyset, & \text{otherwise} \end{array} \right. \end{array}$$

Hence, $\Phi_1(\ulcorner n \urcorner) = \delta^1(\ulcorner n \urcorner)$ for $n \leq 1$.

So it suffices to check $\zeta * \Phi_0 = \zeta * \delta^1 + \epsilon$ for $n \leq 1$.

With



$\zeta * \Phi_0$ is given by $\mathbb{N} \xleftarrow{\text{id}} \mathbb{N} \rightarrow 1$,

On the other hand, with

$$\begin{array}{ccccc}
n_1 + n_2 & & \mathbb{N} & & \\
\uparrow & & \uparrow & & \\
(n_1, n_2) & & \mathbb{N} \times \mathbb{N} & \longleftarrow & \mathbb{N} \\
\downarrow & & \downarrow & & \downarrow \\
(n_1, n_2) & & \mathbb{N} \times \mathbb{N} & \xleftarrow{\text{id} \times \{1\}} & \mathbb{N} \times * \longrightarrow 1
\end{array}$$

$$(n, 1) \longleftarrow n$$

$\zeta * \delta^1$ is given by

$$\mathbb{N} \leftarrow \mathbb{N} \rightarrow 1, \quad n + 1 \leftarrow n,$$

By a map $\text{Grpd} \rightarrow \text{Grpd}$ computed by

$$\begin{array}{ccccccc}
& & ? & \longrightarrow & 1 & & \\
& & \downarrow & & \downarrow \lceil n \rceil & & \\
1 \xrightarrow{\lceil n \rceil} \mathbb{N} & \longmapsto & \mathbb{N} & \longrightarrow & \mathbb{N} & \longmapsto & \begin{cases} \{n-1\} & , n \geq 0 \\ \emptyset & , n = 0 \end{cases}
\end{array}$$

$$x \longmapsto x + 1$$

$$(\zeta * \delta^1)(\lceil n \rceil) = \begin{cases} \{n-1\}, & n \geq 1, \\ \emptyset, & n = 0. \end{cases}$$

Thus:

$$n = 1 : \quad (\zeta * \Phi_0)(\lceil 1 \rceil) = 1, \quad (\zeta * \delta^1)(\lceil 1 \rceil) = \{n-1\} \simeq 1, \quad \varepsilon(\lceil 1 \rceil) = \emptyset,$$

and

$$n = 0 : \quad (\zeta * \Phi_0)(\lceil 0 \rceil) = 1, \quad (\zeta * \delta^1)(\lceil 0 \rceil) = \emptyset, \quad \varepsilon(\lceil 0 \rceil) = 1.$$

Hence for $n \leq 1$,

$$\zeta * \Phi_0 = \zeta * \delta^1 + \varepsilon.$$

Combining the cases $n \geq 2$ and $n \leq 1$, we obtain the objective-level Möbius principle for $\mathbf{N}$.

Remark 9.2. The fact that the Möbius function vanishes for $n \geq 2$ is not a coincidence of alternating sums; rather, at the objective level, the collections of odd and even decompositions for all $n \geq 2$ are themselves equivalent, which leads to their cancellation.

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